

COURSE DESCRIPTION

Course Number	0418-322	Course Title	Computer Architecture
Credit Hours	3 (3:1:3)	Course Coordinator	Dr. Mohammad Almutawa
Compulsory	Yes	Prerequisite	0418-221

Textbook

Title	Computer Organization and Design: The Hardware/Software Interface
Author	Patterson and Hennessy
Publisher	Morgan Kaufmann, Latest Edition

Major Topics Covered in the Course

	Topics	Hours
1	Computer Abstractions: Basics of processors, memory, and performance	2
2	Instructions: Operations in hardware, Decision instructions, MIPS Addressing, Parallelism and Instructions	3
3	Processor: Single and multiple cycle implementations and Parallelism, Pipelining	4
4	Pipelining: Data Hazard and Resolution, Control Hazard and Resolution, Branch Prediction, and Exceptions	8
5	Memory Hierarchy	8
6	Input/output	5
7	Parallel Processors	6
8	Graphical Processing Unit	3
9	Exams	3
	Total	42

Grading

15-20%	Midterm 1
15-20%	Midterm 2
15-20%	Lab
0-15%	Homework/Quizzes
40%	Final

Course outcomes

Students should be able to:

1. Understand the concepts of computer abstraction, design, performance, single and multiple cycle implementations, pipelining, parallelism, various processor architectures.
2. Understand why a memory hierarchy is necessary to reduce the effective memory latency and describe the various ways of organizing cache memory and appreciate the cost-performance tradeoffs for each arrangement.
3. Analyze the performance of I/O devices and how they interfacing with processor, memory and operating system.
4. Understand how multicore, multiprocessors and clusters increase the performance and the needs to express algorithms in suitable forms.
5. Understand the concepts of ISA design, MIPS ISA, multimedia value representation, and special-purpose graphics processor and performance.

Relationship between Course Outcomes and Student Outcomes

Course Outcomes	Student Outcomes					
	1	2	3	4	5	6
1	H					
2	M					
3	M					
4	M					
5	M					